3.4 Vectors and Dot Products

Name _____ Period ____ Date _____

Pre-Calculus 3.4 · Student sheet

I can…

- find the dot product of two vectors and use its properties.
- find the angle between two vectors.
- decide whether two vectors are orthogonal.

Vocabulary · fill in the missing word

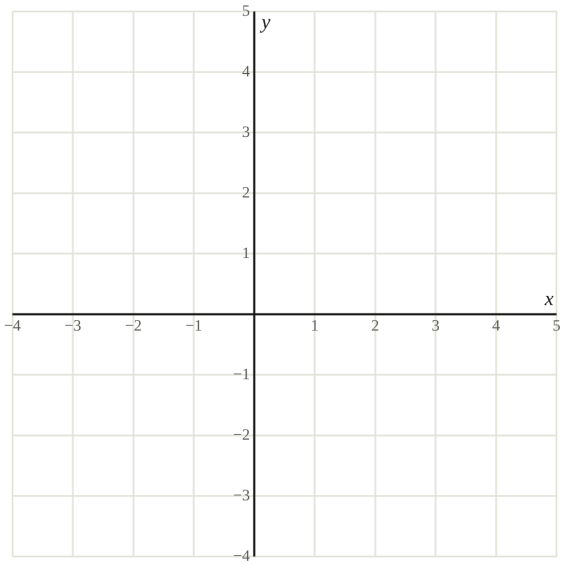
Dot product — $u \cdot v = u_1v_1 + u_2v_2$ (a number, not a _____)

Angle between vectors — $\cos\theta =$ _____, $0 \le \theta \le \pi$

Orthogonal — _____: $u \cdot v = 0$

Part 1 · Perpendicular test

Draw each pair of vectors from the origin. Measure the angle between them, then compute $u_1v_1 + u_2v_2$.



u	v	Angle (measured)	$u_1v_1 + u_2v_2$
⟨2, 1⟩	⟨−1, 2⟩		
⟨3, 4⟩	⟨4, −3⟩		
⟨1, 2⟩	⟨2, 1⟩		
⟨2, 3⟩	⟨−3, 1⟩		

Pattern: 0 means _____, positive means _____, negative means _____

Part 2 · Notes: fill in the blanks

Dot product: $u \cdot v =$ _____. The answer is a _____.

$u \cdot u =$ _____

Angle between u and v: $\cos\theta =$ _____

u and v are **orthogonal** when _____.

Part 3 · Examples (work them with the class)

Example 1

Find each dot product.

a. $\langle 4, 5\rangle \cdot \langle 2, -3\rangle$ b. $\langle 0, 3\rangle \cdot \langle 4, -2\rangle$

Example 2

Let $u = \langle -1, 3\rangle$, $v = \langle 2, -4\rangle$, and $w = \langle 1, -2\rangle$. Find each quantity. Is it a vector or a scalar?

a. $(u \cdot v)w$ b. $u \cdot 2v$ c. $\|u\|$

Example 3

Find the angle between $u = \langle 4, 3\rangle$ and $v = \langle 3, 5\rangle$.

Example 4

Are $u = \langle 2, -3\rangle$ and $v = \langle 6, 4\rangle$ orthogonal?

Example 5

Find $u \cdot v$ when $\|u\| = 10$, $\|v\| = 7$, and the angle between them is $\frac{2\pi}{3}$.

Example 6

Use vectors to find the interior angles of the triangle with vertices A(1, 2), B(3, 4), and C(2, 5).

Part 4 · Partner practice

Partner 1

Find the angle between $\langle 1, 2\rangle$ and $\langle 3, -1\rangle$.

Partner 2

Find k so that $\langle 2, k\rangle$ and $\langle 3, -6\rangle$ are orthogonal.